

空中机器人离散型壳体自主无人倾斜砌筑

Autonomous Unmanned Inclined Masonry of Discrete Shells Using Aerial Robots

——设计畅想·未来无人机砌筑思考

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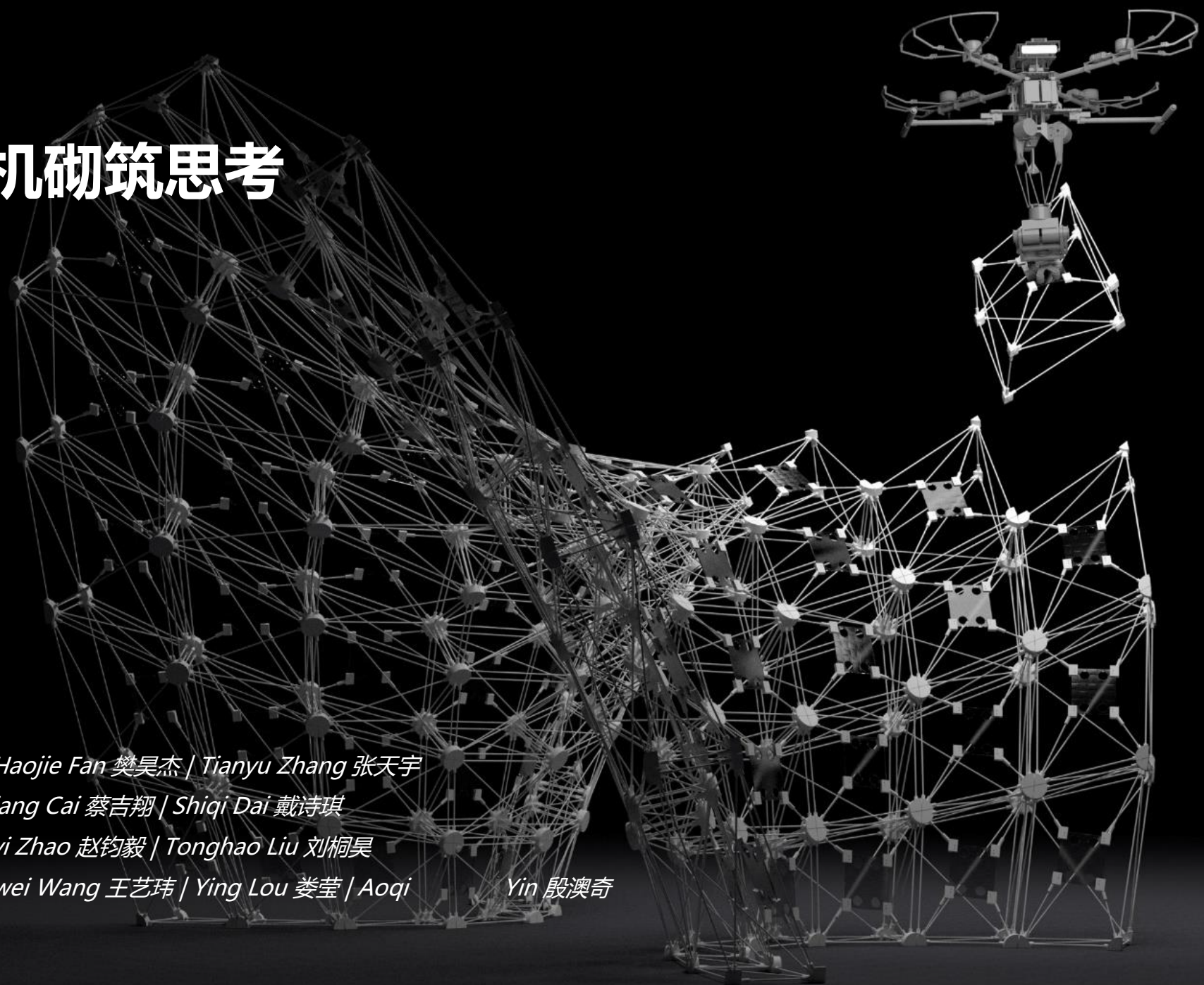
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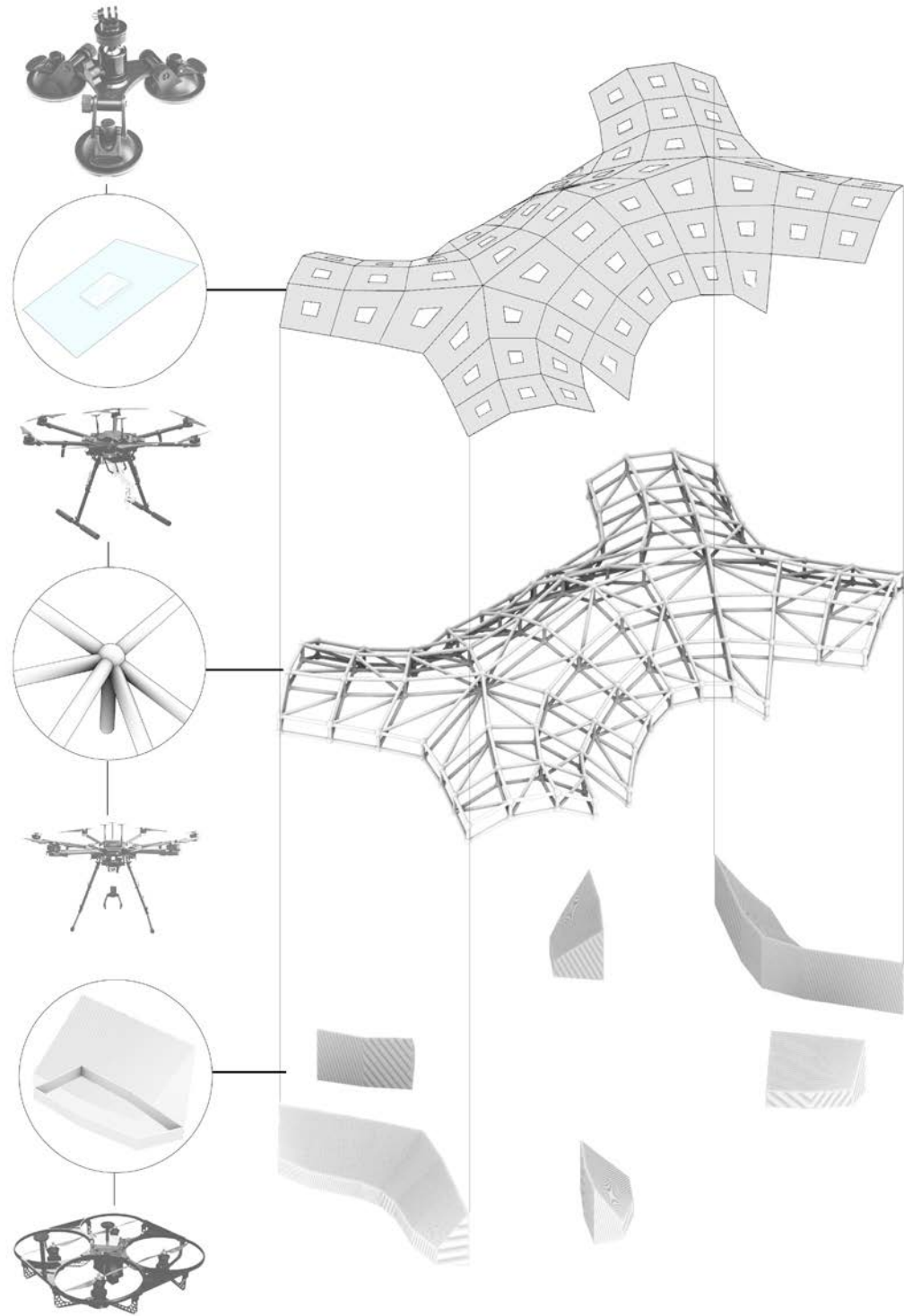
多无人机协同过程 Multi-drone collaborative process

STEP3: 杆件抓取无人机
对顶部玻璃面板进行放置
The rod grab drone is
used to place the top
glass panel

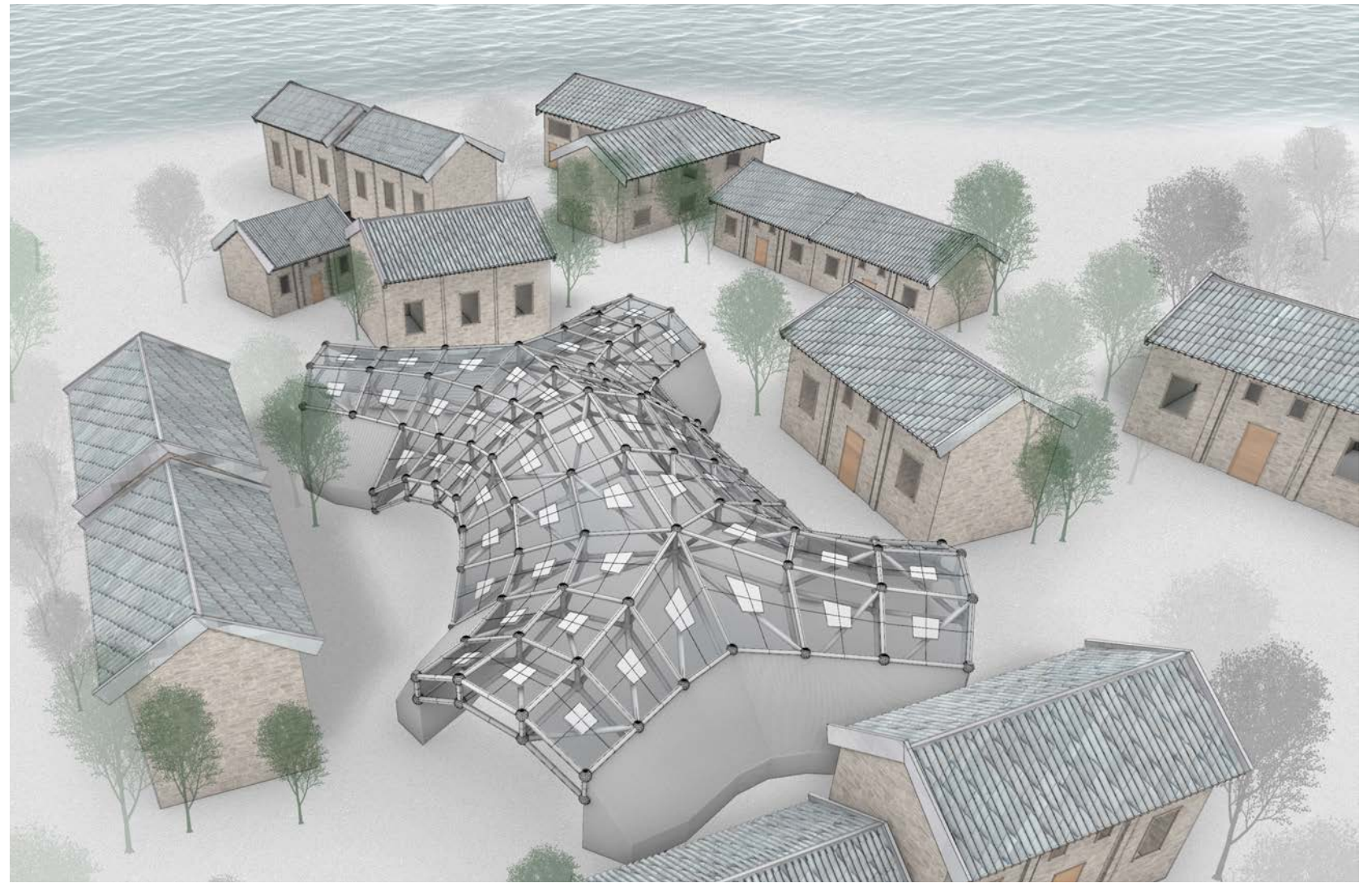
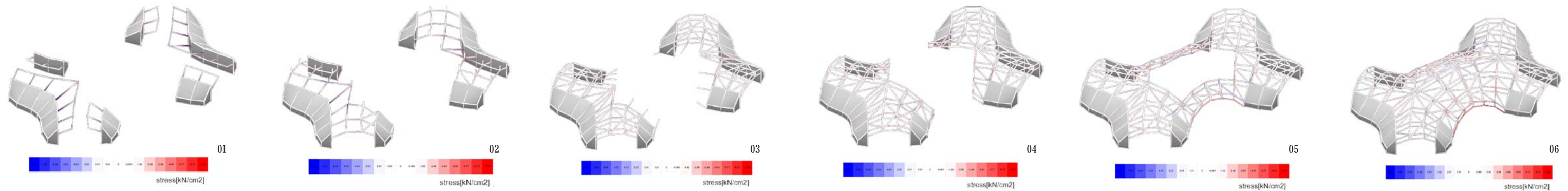
STEP2: 焊接无人机对金属
单元杆件进行组装
Welding drones
assemble metal unit
rods

STEP2: 组装机器人对杆
件进行自动化装配
Assembly robots
automate the assembly
of rods

STEP1: 3D 打印无人机进
行混凝土基础打印
3D printing drones
print concrete
foundations



搭建过程及有限元分析 (FEA) Construction process and finite element analysis



Weizhou Island multi-air robot collaboration workflow

Designer: Zhao Junyi , Yan Xiheng , Liu Haotong

Due to the large number of coral rocks in the underground layer of Weizhou Island, the government has strictly prohibited large vehicles from entering in order to protect the natural environment, which has caused certain difficulties in the construction of local buildings. The design explores a workflow of automated building construction using drones for multi-machine collaboration to solve the construction problems of Weizhou Island.

形体生成 Form generation

